

Supply Chain Operations & Logistics

Cross-Functional Executive Summary Report

Executive Summary

This report provides leadership with a unified view of end-to-end supply chain health, identifying operational bottlenecks from inbound procurement to outbound fulfillment and reverse logistics. By extracting and normalizing over 500,000 rows of retail data, this analysis bridges the gap between raw data and actionable operational strategy.

Key Operational Findings

1. Inbound Operations (Supplier Reliability)

Overall network lead time variance averages **1.00 days late**. Granular OTIF (On-Time In-Full) tracking reveals clear opportunities to consolidate purchase orders with top-performing partners (e.g., Global Components) to reduce buffer stock requirements.

2. Warehouse Management (Inventory Integrity)

Cycle count audits revealed isolated SKUs falling below the 95% record accuracy threshold. By implementing automated variance triggers, physical inventory teams can pivot from time-consuming full-facility counts to targeted daily audits on high-risk items.

3. Outbound & Reverse Logistics

After normalizing outbound flow data and filtering out administrative write-offs (such as "missing" or "Zebra invcing error"), genuine product returns were isolated. The highest physical return volumes are concentrated in a specific subset of decorative SKUs.

Strategic Recommendations

- **Targeted Vendor Renegotiations:** Leverage the supplier scorecard data during the next quarter's contract renewals to penalize or replace vendors consistently breaching the 5% defect rate or missing delivery windows.

- **Dynamic Cycle Counting:** Shift warehouse labor hours toward daily audits of the specific SKUs glowing red on the accuracy matrix to aggressively reduce inventory shrinkage.
- **Cross-Departmental Quality Review:** Initiate a joint review with the Procurement and Quality Assurance teams focusing on the top returned items to determine if the root cause is poor supplier manufacturing or damage during outbound transit.

Technical Architecture

Discipline	Technologies Applied
Database Engineering	MySQL (Data extraction, normalization, handling missing variables)
Business Intelligence	Power BI (Cross-filtering, directional flow modeling, automated threshold alerting)
Data Processing	DAX Measures, Power Query
Data Analysis	Microsoft Excel (PivotTables, conditional formatting, calculated variance tracking)